FEM for 1D problems

Numerical solution

- Element-wise computation of $\underline{K}$ and $\underline{r}$

Motivation

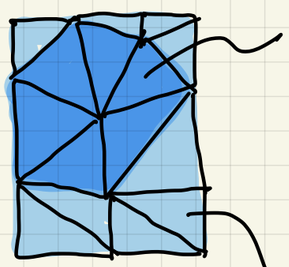
We computed $\underline{K}$ and $\underline{r}$ using

$$K_{ij} = a(\varphi_i, \varphi_j)$$
$$r_i = b(\varphi_i)$$

} Node-wise!

For practical applications (especially in 2D or 3D) this is way too complicated.

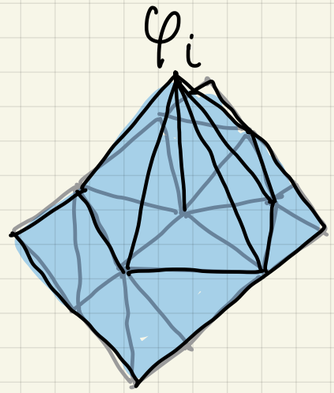
In 2D



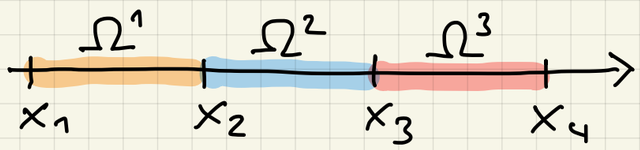
Area where $\varphi_i(x_1, x_2) \neq 0$

Complicated!

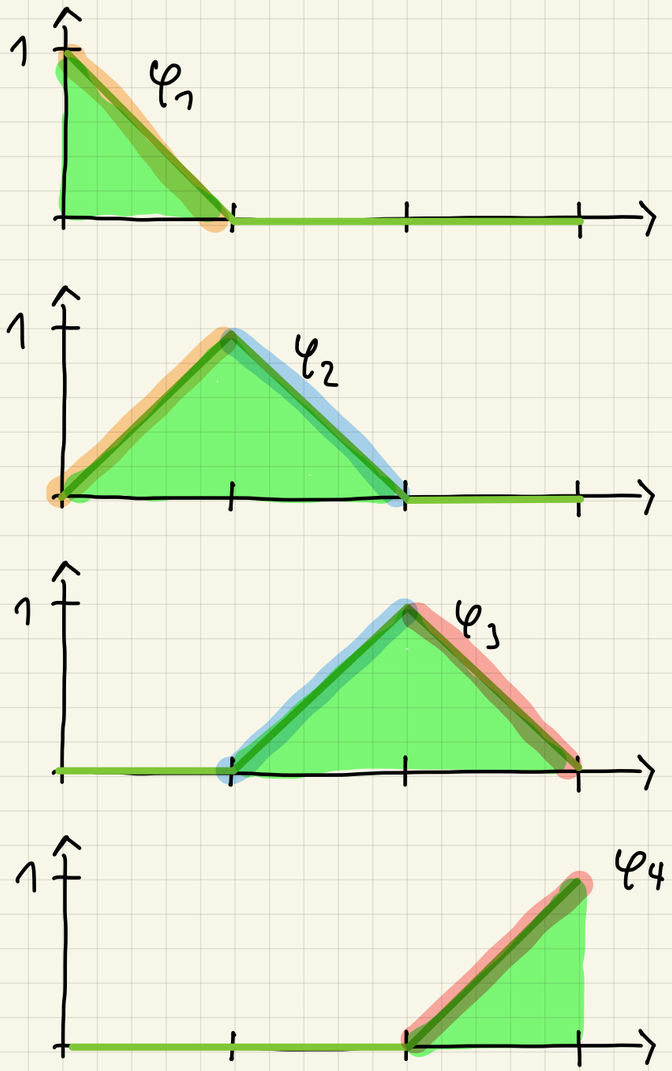
Non-uniform elements



Basis functions φ_i (global) and shape functions N_k^e (local)



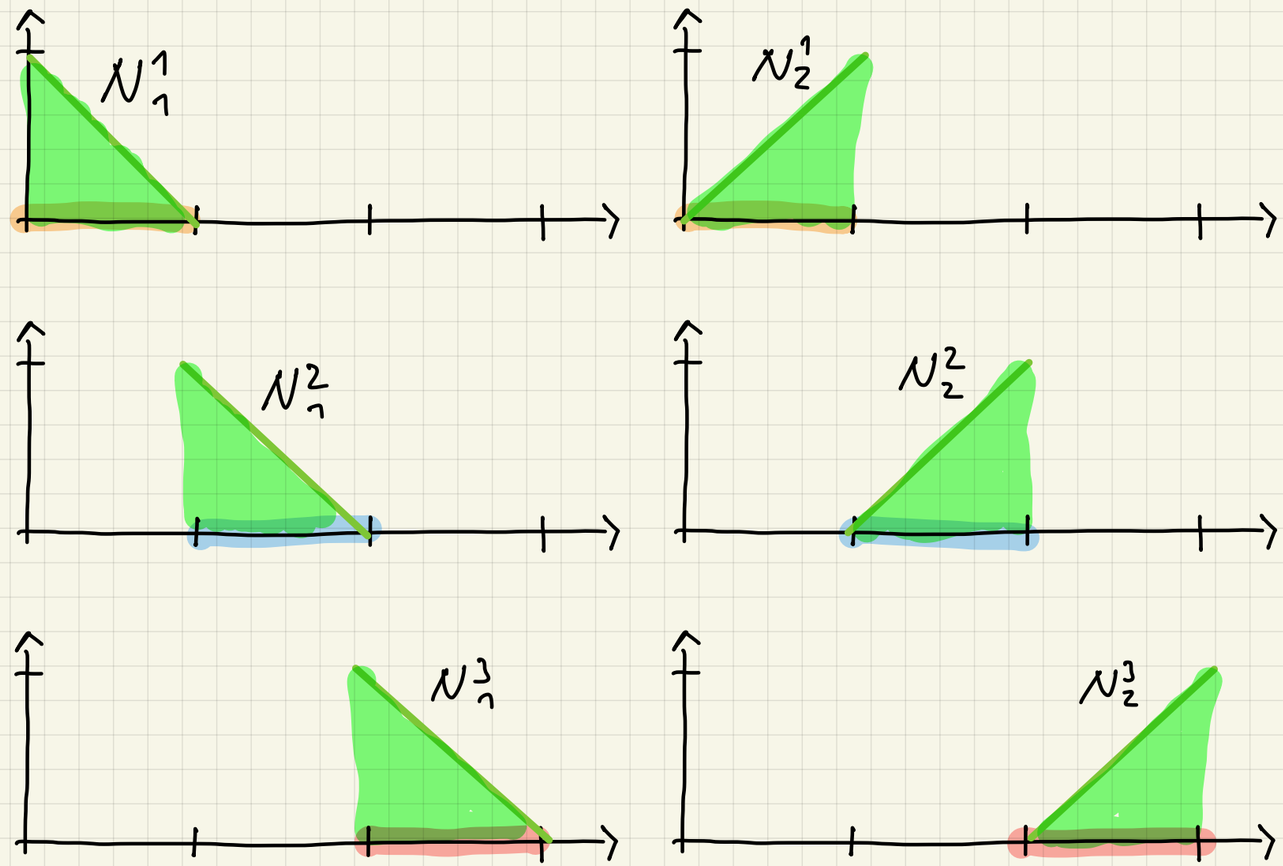
Basis functions



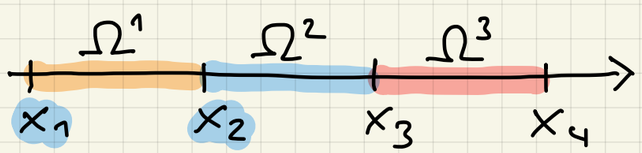
$\longleftrightarrow$

Shape functions

$N_k^e : \Omega^e \to \mathbb{R}, \; k = 1,2$

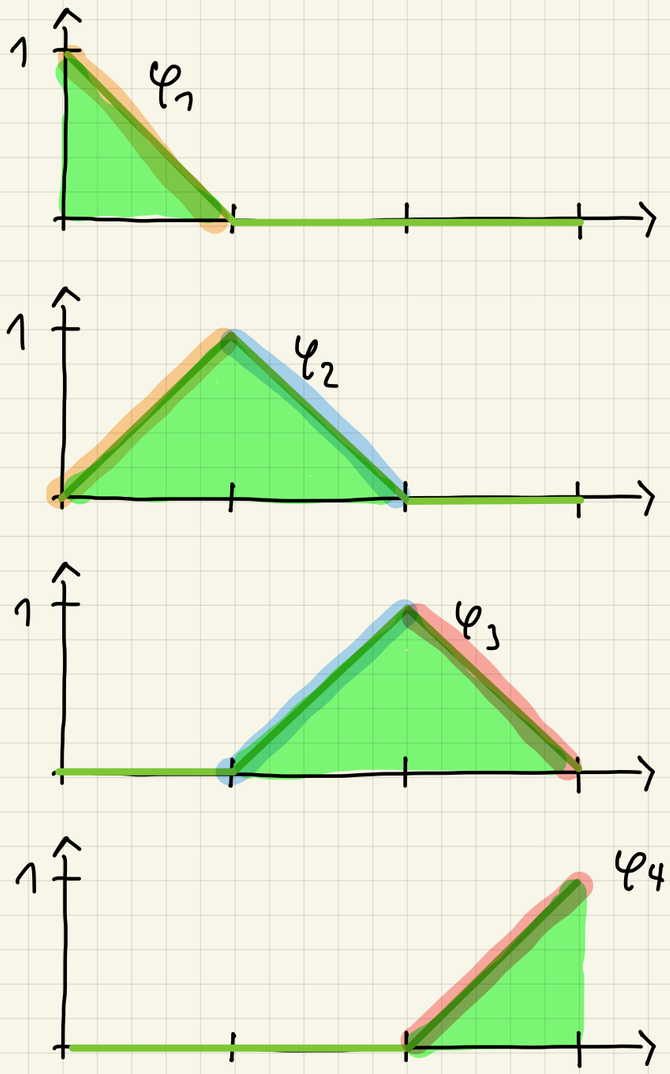


Basis functions φ_i (global) and shape functions N_k^e (local)



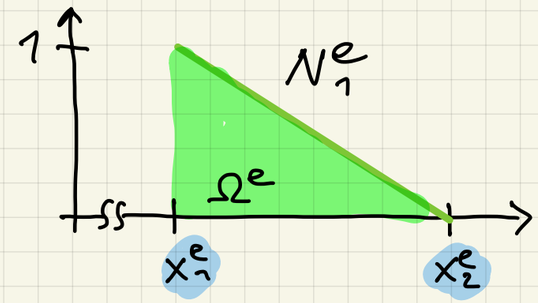
We always have the same kind of functions on one element!

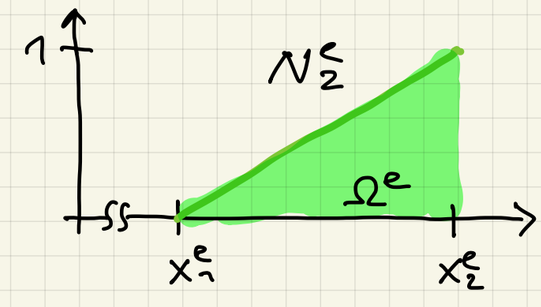
Basis functions



Shape functions

$N_k^e : \Omega^e \to \mathbb{R}, \quad k = 1,2$





Connection

On Ω^1

$$\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} N_1^1 \\ N_2^1 \end{bmatrix}$$

On Ω^2

$$\begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \\ \varphi_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} N_1^2 \\ N_2^2 \end{bmatrix}$$

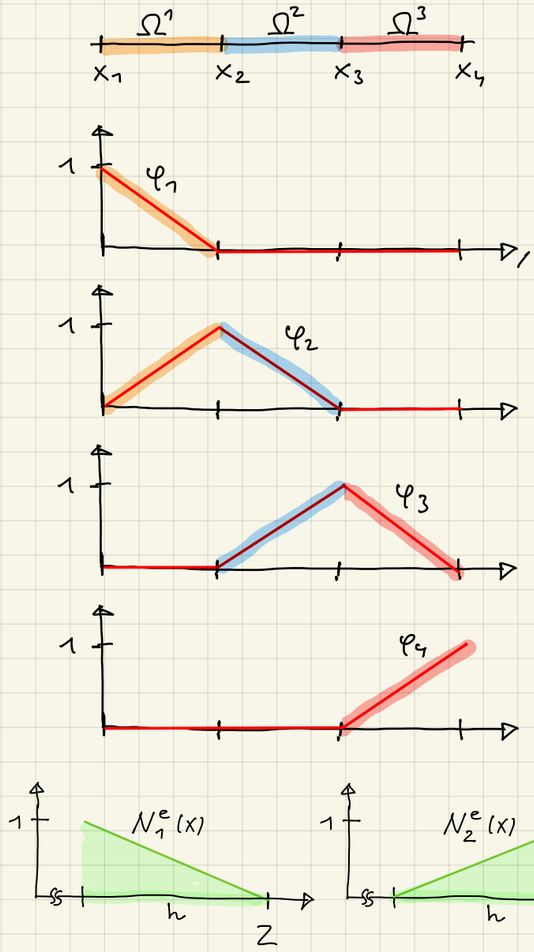
Transpose of assembly matrix (by convention)

Assembly matrices

$$\underline{\underline{A}}^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{matrix} N_1^1 \\ N_2^1 \end{matrix} \qquad \underline{\underline{I}}^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\underline{\underline{A}}^2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{matrix} N_1^2 \\ N_2^2 \end{matrix} \qquad \underline{\underline{I}}^2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\underline{\underline{A}}^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} N_1^3 \\ N_2^3 \end{matrix} \qquad \underline{\underline{I}}^3 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$



$$A^e_{ki} = \begin{cases} 1 & \text{if } i = I^e_k \\ 0 & \text{otherwise} \end{cases}$$

$k = 1,2 \quad , \quad i = 1, \dots, N$

Note: Assembly matrices not required for implementation

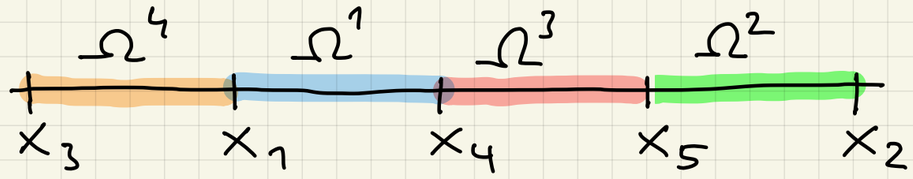
Connection of global and local functions on Ω^e

$$\varphi_i = \sum_{k=1}^{2} A_{ki} \cdot N^e_k$$

$$\underline{y} = \underline{\underline{A}}\,\underline{x} \qquad y_i = \sum_{k=1}^{m} A_{ik} \cdot x_k$$

$$\underline{y} = \underline{\underline{A}}^T \underline{x} \qquad y_i = \sum_{k=1}^{n} A_{ki} \cdot x_k$$

Assignment Ü4/1.1



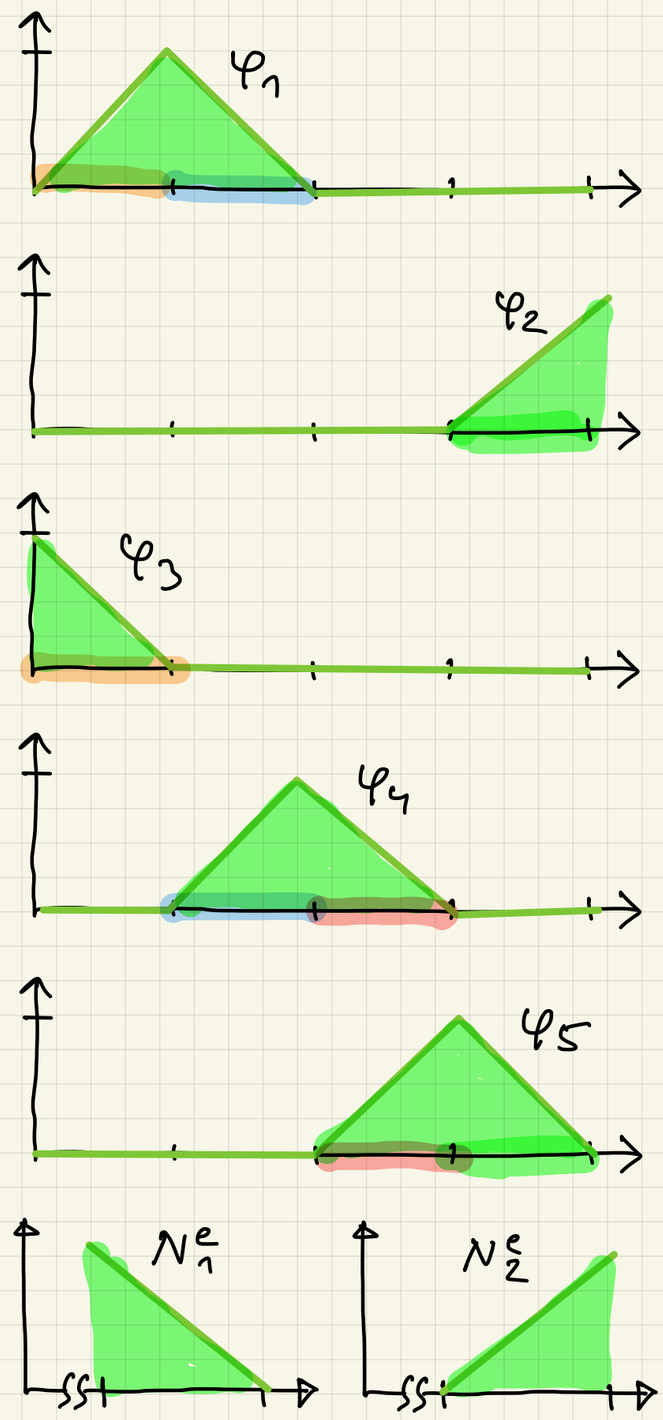
$$\underline{A}^1 = \begin{array}{c} \\ \\ \end{array} \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \quad 5 \\ \left[\begin{array}{ccccc} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right] \end{array} \begin{array}{c} \\ 1 \\ 2 \end{array} \qquad \underline{I}^1 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$\underline{A}^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \qquad \underline{I}^2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$\underline{A}^3 = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \underline{I}^3 = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$\underline{A}^4 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \qquad \underline{I}^4 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

→ Node numbers indicate position of 1 !



Assembly of global stiffness matrix

Element bilinear form

$$a^e(u,\delta u) = EA\int_{x_1^e}^{x_2^e} u'\,\delta u'\,dx + c\cdot\int_{x_1^e}^{x_2^e} u\cdot\delta u\,dx$$

(No S!)

Global stiffness matrix

$$K_{ij} = a(\varphi_i,\varphi_j) = \sum_{e=1}^{Ne} a^e(\varphi_i,\varphi_j) + K_{ij}^s = \sum_{e=1}^{Ne} a^e\Big(\sum_{k=1}^{2} A_{ik}^e\cdot N_k^e, \sum_{L=1}^{2} A_{jL}^e\cdot N_L^e\Big) + K_{ij}^s$$

$$= \sum_{e=1}^{Ne}\sum_{k=1}^{2}\sum_{L=1}^{2} A_{ik}^e\cdot a^e(N_k^e,N_L^e)\cdot A_{jL}^e + K_{ij}^s$$

Element stiffness matrix

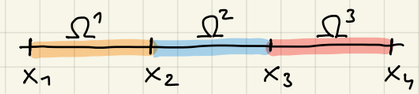
$$K_{kl}^e = a^e(N_k^e,N_l^e)$$

In matrix notation

$$\underline{K} = \sum_{e=1}^{Ne} (\underline{A}^e)^T\,\underline{K}^e\,\underline{A}^e + \underline{K}^s$$

$$\begin{bmatrix} N\times 2 \end{bmatrix}\begin{bmatrix} 2\times 2 \end{bmatrix}\begin{bmatrix} \quad 2\times N \quad \end{bmatrix} = \begin{bmatrix} \quad N\times N \quad \end{bmatrix}$$

Assembly of global stiffnes matrix (example)



$$\underline{\underline{K}} = \sum_{e=1}^{3} (\underline{\underline{A}}^e)^T \underline{\underline{K}}^e \underline{\underline{A}}^e = \begin{bmatrix} K^1_{11} & K^1_{12} & 0 & 0 \\ K^1_{12} & K^1_{22}+K^2_{11} & K^2_{12} & 0 \\ 0 & K^2_{12} & K^2_{22}+K^3_{11} & K^3_{12} \\ 0 & 0 & K^3_{12} & K^3_{22} \end{bmatrix}$$

$$\underline{\underline{A}}^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \qquad \underline{\underline{A}}^2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad \underline{\underline{A}}^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Ω^1 :

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} K^1_{11} & K^1_{12} \\ K^1_{12} & K^1_{22} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} K^1_{11} & K^1_{12} \\ K^1_{12} & K^1_{22} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} K^1_{11} & K^1_{12} & 0 & 0 \\ K^1_{12} & K^1_{22} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \leftarrow \leftarrow$$

$$\underline{\underline{I}}^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Ω^2 :

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} K^2_{11} & K^2_{12} \\ K^2_{12} & K^2_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ K^2_{11} & K^2_{12} \\ K^2_{12} & K^2_{22} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & K^2_{11} & K^2_{12} & 0 \\ 0 & K^2_{12} & K^2_{22} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \leftarrow \leftarrow$$

$$\underline{\underline{I}}^2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Ω^3 :

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} K^3_{11} & K^3_{12} \\ K^3_{12} & K^3_{22} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ K^3_{11} & K^3_{12} \\ K^3_{12} & K^3_{22} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & K^3_{11} & K^3_{12} \\ 0 & 0 & K^3_{12} & K^3_{22} \end{bmatrix} \leftarrow \leftarrow$$

$$\underline{\underline{I}}^3 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

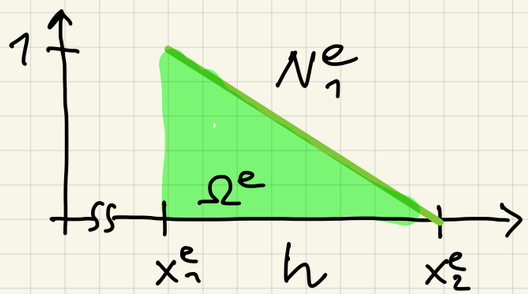
Übungsblatt 4

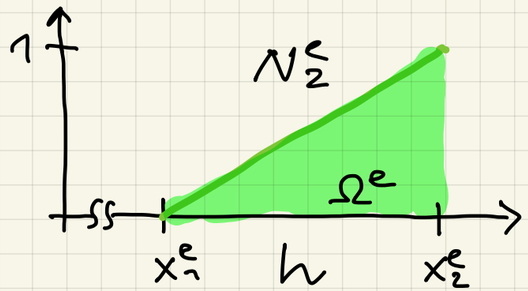
Aufgabe 2

Computation of element stiffness matrix

$$K^e_{kl} = a^e(N^e_k, N^e_l) = EA\int_{x^e_1}^{x^e_2} N^{e\,\prime}_k \cdot N^{e\,\prime}_l \, dx + C \cdot \int_{x^e_1}^{x^e_2} N^e_k \cdot N^e_l \, dx$$

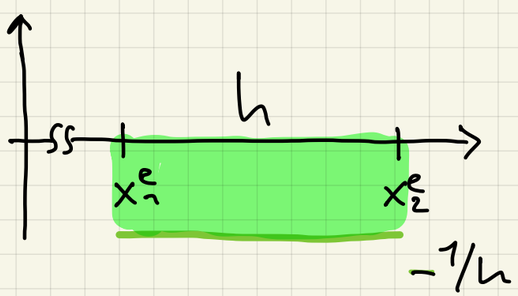
Integrals

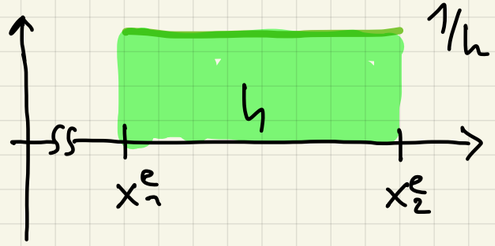




Ü3/1.1

$$\int_{x^e_1}^{x^e_2} N^e_k \cdot N^e_l \, dx = \begin{cases} h/3 & \text{for } k=l \\ h/6 & \text{otherwise} \end{cases}$$





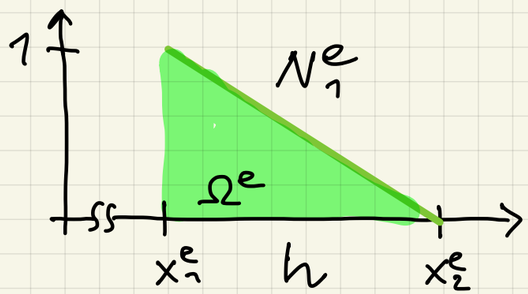
$$\int_{x^e_1}^{x^e_2} N^{e\,\prime}_k \cdot N^{e\,\prime}_l \, dx = \begin{cases} 1/h & \text{for } k=l \\ -1/h & \text{otherwise} \end{cases}$$

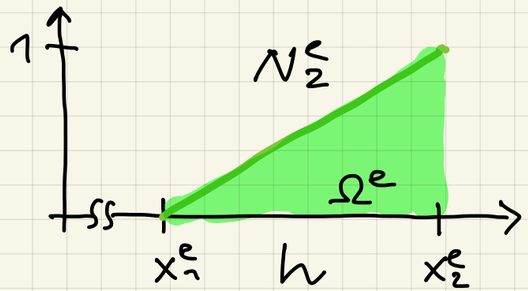
$$\underline{K}^e = \frac{EA}{h}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} + \frac{C \cdot h}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Computation of element load vector

$$r_k^e = b^e(N_k^e) = n \cdot \int_{x_1^e}^{x_2^e} N_k^e \, dx$$

Integrals





$$\int_{x_1^e}^{x_2^e} N_k^e \, dx = \frac{h}{2}$$

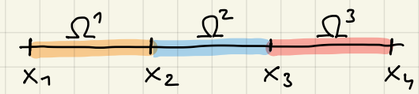
$$\underline{r}^e = \frac{n \cdot h}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Assembly of global load vector

TODO

Ende Aufgabe 2.2

Assembly of global stiffnes matrix (example)



$$\underline{\underline{K}} = \sum_{e=1}^{3} (\underline{\underline{A}}^e)^T \underline{\underline{K}}^e \underline{\underline{A}}^e = \begin{bmatrix} K^1_{11} & K^1_{12} & 0 & 0 \\ K^1_{12} & K^1_{22}+K^2_{11} & K^2_{12} & 0 \\ 0 & K^2_{12} & K^2_{22}+K^3_{11} & K^3_{12} \\ 0 & 0 & K^3_{12} & K^3_{22} \end{bmatrix}$$

$$\underline{\underline{A}}^1 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \qquad \underline{\underline{A}}^2 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad \underline{\underline{A}}^3 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Ω^1 :

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} K^1_{11} & K^1_{12} \\ K^1_{12} & K^1_{22} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} K^1_{11} & K^1_{12} \\ K^1_{12} & K^1_{22} \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} K^1_{11} & K^1_{12} & 0 & 0 \\ K^1_{12} & K^1_{22} & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \leftarrow \leftarrow$$

$$\underline{\underline{I}}^1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Ω^2 :

$$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} K^2_{11} & K^2_{12} \\ K^2_{12} & K^2_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ K^2_{11} & K^2_{12} \\ K^2_{12} & K^2_{22} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & K^2_{11} & K^2_{12} & 0 \\ 0 & K^2_{12} & K^2_{22} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \leftarrow \leftarrow$$

$$\underline{\underline{I}}^2 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Ω^3 :

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} K^3_{11} & K^3_{12} \\ K^3_{12} & K^3_{22} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ K^3_{11} & K^3_{12} \\ K^3_{12} & K^3_{22} \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & K^3_{11} & K^3_{12} \\ 0 & 0 & K^3_{12} & K^3_{22} \end{bmatrix} \leftarrow \leftarrow$$

$$\underline{\underline{I}}^3 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Implementation in Matlab

The product $(\underline{A}^e)^T \underline{K}^e \underline{A}^e$

We have seen: Multiplication with $(\underline{A}^e)^T$ and $\underline{A}^e$ rearranges the entries of the local matrix $\underline{K}^e$ into the global matrix $\underline{K}$.
We can derive the scheme:

For each element e and $k,l = 1,2$

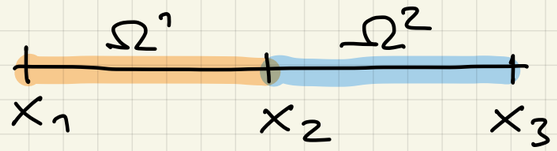
$$K_{I_k^e I_l^e} \leftarrow K_{I_k^e I_l^e} + K^e_{kl}$$

In Matlab, this can easily obtained using an index-vector.

→ Assignment Ü4/3.1

Finite element data structure

Pile foundation with 2 elements




```
S = struct();
S.nodes    = [0  10  20];
S.elements = [1  2 ; 2  3];
```

We store

- Node positions: Column j contains the coordinates of node j
- Elements: Column e contains node numbers of element e

→ Assignment Ü4/3.2

Functions

Element stiffness matrix

```
function keFunc = pileKe(EA,C)
    function ke = computeKe(x)
        ...
    end
    keFunc = @computeKe;
end
```

Node coordinates of element

Usage

```
keFunc = pileKe(10  ,20);
Ke     = keFunc([0,5]);
```

Element load vector

function reFunc = pileRe(n)

Usage : See above

Assembly of linear System

[K,r] = assembleKr(S)

...

end

(Not more than 30 lines of code) ←

Usage

...

S.keFunc = pileKe(p.EA, p.C);

S.reFunc = pileRe(p.n);

[K,r] = assembleKr(S);

Steps:

- $\underline{\underline{K}}$ as $N \times N$ matrix of zeros
- $\underline{r}$ as vector of N zeros
- Loop over all elements
  - Get element node numbers $\underline{I}^e$
  - Collect node coordinates $\underline{x}^e$
  - Compute $\underline{\underline{k}}^e$ and $\underline{r}^e$
  - Add $\underline{\underline{k}}^e$ to $\underline{\underline{K}}$ and $\underline{r}^e$ to $\underline{r}$

Note: Use

ke = S.keFunc(xe);

re = S.reFunc(xe);

Good news

- assemble Kr can also be used for 2D and 3D problems
- Required changes
  - mesh
  - Functions to compute $\underline{k}^e$ and $\underline{r}^e$

Aufgaben 3.3 und 3.4